

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	1.53%	1.45%	20.35%	2.47%
Meat MCal Target	83,622.96	79,526.57	1,114,078.19	135,322.29
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	66.27	0.00	507.60	63.45
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	1,316.74	483.07	1,484.40	2,130.80
MCals from meat	83622.96	79526.57	1114078.19	135322.29
Max Utilized Mcals allowed from Dairy	87260.19	0.00	753479.36	135199.36
MCals actually Utilized from Dairy	87260.19	0.00	753479.36	135199.36
Percent of diet from their meat	1.53%	1.45%	20.35%	2.47%
Percent of Diet from their Dairy	1.59%	0.00%	13.76%	2.47%
Kg Fleece	9,950.72	0.00	0.00	0.00
1 Year+ (age) skins	489.55	18.66	285.12	792.03
Yearly Offspring skins	2,396.12	569.69	2,186.37	3,877.14
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	2,351.32	88.23	2,182.94	3,805.00
Males Step 1	658.37	241.54	742.20	1,065.40
Females Step 1	658.37	241.54	742.20	1,065.40
Replacement Mothers per year	470.56	17.95	273.05	761.30
Breeding Males Required	123.44	4.63	114.60	199.76
Additional Mothers	1,927.47	15.82	1,032.31	3,118.46
Males Step 2	539.69	43.31	350.99	873.17
Females 2	539.69	43.31	350.99	873.17
Total Mothers	2,821.88	106.18	2,455.99	4,566.30
Total Males	1,198.06	284.84	1,093.18	1,938.57
Total Females	118.68	198.23	391.21	192.23
Total Offspring	2,396.12	569.69	2,186.37	3,877.14
Total Animals	4,262.06	593.88	4,054.99	6,896.87
Units per Hectare	1.00			
Total Adults per Hectare	1.71	0.26	0.42	1.71
Total Offspring per Hectare	0.84	1.15	0.25	0.84
	1.05%		12.78%	1.70%
Total Adults	4,873	127	3,603	7,885
Total Offspring	2,396	570	2,186	3,877
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	425.76	0.00	269.62	689.17
Other	38,337.95	0.00	0.00	62,034.19
MCals Dry Mothers	497,286.08	0.00	570,792.49	804,946.48
total Maintenance MCals	535,624.03	0.00	570,792.49	866,980.67

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	1,426,890	260,091	3,659,977	2,308,835
Lactating	2,722,234	92,460	3,935,459	4,404,815
Maintenance	535,624	0	570,792	866,981
Adult Males	144,183	0	242,617	233,323
Offspring	657,735	572,537	8,002,100	1,064,274
TOTAL	5,486,666.6	925,089.2	16,410,946.2	8,878,227.7

Animal Diet Breakdown				
	Sheep	Pig	Cow	Goat
	Crop Barley	Barley	Barley	Barley
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0
MCal kg ⁻¹				
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0
MCal kg ⁻¹				
Days	90.0	90.0	90.0	90.0
Number of Animals	2,396.1	569.7	2,186.4	3,877.1
MCals from Straw/Chaff	306224.4	0.0	1384793.7	495498.1
MCals from Grain	485214.6	115362.0	885478.3	785120.2
MCals from other feed	791439.0	115362.0	2270272.0	1280618.3

Diet Calculations		
Plants Mcal	2,856,561.089	
Meat Mcal	1,642,500	
Dairy	975,938.9114293	
	Perfect	
Plants %	52.17%	
Meat %	30.00%	
Dairy %	17.83%	
check	Perfect	
Percent		
Domestic Meat	25.80%	
Dairy	17.83%	
Wild Meat	4.20%	
Crops	46.96%	
Wild Plants	5.22%	
check	Perfect	
new dairy	975938.91	
potential dairy < limit	TRUE	
new limit	70.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	1,412,550.000	
Dairy	975,938.911	
Wild Meat	229950.000	
Crops	2,570,904.980	
Wild Plants	285,656.109	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd Males for Slaughter Females for Slaughter	2822	106	2456	4566				
	1198	285	1093	1939				
	119	198	391	192				
Offspring available for Consumption	2396	570	2186	3877				
Gestation Req'd	784117	220547	2484879	1268892				
Lactation Req'd	1495945	78403	2671913	2420804				
Maintenance Req'd	1779075	0	731001	2878759				
Adult Males	144183	0	242617	233323				
Offspring Required	787126	623809	9576284	1273639				
Energy Required	4990446	922759	15706694	8075417				
Ewe Fleece Kg Yr-1	6349							
Ram Fleece Kg year ⁻¹	247							
Juvenile Fleece Kg Yr-1	3355							
Kg Fleece Year-1	9951	0	0	0				